

- 18** Object X is at a small height h above the surface of the Earth and then falls from rest to the surface.

The speed of object X when it reaches the surface of the Earth is v_X .

Object Y is at a small height $2h$ above the surface of the Moon and then falls from rest to the surface.

The speed of object Y when it reaches the surface of the Moon is v_Y .

The acceleration of free fall on the Moon is 1.6 m s^{-2} .

The drag force acting on both objects is negligible.

What is the ratio $\frac{v_X}{v_Y}$?

A 1.0

B 1.8

C 2.5

D 3.1